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PENETRATION TESTING REPORT (SAMPLE)

Footprinting & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Detail
Pentester Name	[Ibrahim Idris Ibrahim]
Program/Batch	B082 - Networkwalks
Date	[12/11/2026]
Modules Completed	W2-PM1 (Footprinting with Multiple Kali Tools), W2-PM5 (Zenmap Network Scanning), W2-PM-FINAL (Report)

1. Executive Summary

This report documents the reconnaissance and network scanning phases conducted as part of the Networkwalks Cybersecurity Internship (Batch B082). The objective was to gather publicly available information about `networkwalks.com` using multiple Kali Linux footprinting tools, and to perform host discovery and port scanning on the tester's local lab network using Zenmap.

Key findings from the footprinting phase include the identification of the website's technology stack, DNS infrastructure, and the presence of a Web Application Firewall. From the network scanning phase, multiple live hosts were discovered on the local subnet, with specific open ports and services identified on the target host.

All activities were conducted within the authorized scope defined in the engagement letter. No exploitation, brute-force, or disruptive testing was performed.

2. Scope & Authorization

This engagement was conducted under the written authorization provided by Networkwalks (Client) for Batch B082. The authorized scope included:

- **Target 1:** `networkwalks.com` — the Client's public website and associated DNS records (passive/light footprinting only).

- **Target 2:** The tester's own local area network (LAN) — owned and controlled by the tester (network scanning only).

Out of Scope: Any system, domain, IP address, or network not listed above. Exploitation, denial-of-service, brute-force, and social engineering were strictly prohibited.

Authorization Reference: See Appendix A for the signed authorization letter.

3. Methodology & Tools Used

The assessment followed a standard penetration testing methodology covering Phase 1 (Reconnaissance) and Phase 2 (Scanning).

3.1 Footprinting Tools (W2-PM1)

Tool	Purpose
WHOIS	Queried domain registration databases to identify registrar, creation date, and name servers.
WhatWeb	Fingerprinted the website's CMS, server software, and detected technologies.
NSLookup	Resolved DNS records (A and MX) to identify IP addresses and mail servers.
Curl	Inspected HTTP response headers to identify server software and security headers.
Wafw00f	Detected the presence and type of Web Application Firewall protecting the site.
DNSRecon	Performed DNS enumeration to discover additional records and subdomains.

3.2 Network Scanning Tools (W2-PM5)

Tool	Purpose
Zenmap (Ping Scan)	Discovered live hosts on the local subnet without port scanning.
Zenmap (Quick Scan)	Performed a fast scan of the most common ports on the target host.
Zenmap (Regular Scan)	Scanned the target host for open ports and service versions.
Zenmap (Intense Scan)	Performed comprehensive scanning including OS detection and version detection.

4. Findings – Footprinting (networkwalks.com)

4.1 WHOIS Lookup

Screenshot: `01_whois.png`

Observation: - **Registrar:** [e.g., GoDaddy, Namecheap, etc.] - **Creation Date:** [e.g., 2015-06-12] - **Name Servers:** [e.g., ns1.cloudflare.com, ns2.cloudflare.com] - **Registrant Info:** [Redacted / Available]

4.2 WhatWeb Fingerprinting

Screenshot: `02_whatweb.png`

Observation: - **CMS:** [e.g., WordPress 6.x] - **Server:** [e.g., Apache/2.4.x] - **Country:** [e.g., US] - **Detected Plugins:** [e.g., Yoast SEO, Contact Form 7]

4.3 NSLookup (A and MX Records)

Screenshots: `03_nslookup_A.png`, `04_nslookup_MX.png`

Observation: - **A Record (IP):** [e.g., 104.21.x.x] - **MX Records:** [e.g., mail.networkwalks.com, alt1.aspmx.l.google.com] - **Mail Provider:** [e.g., Google Workspace]

4.4 Curl – HTTP Headers

Screenshot: `05_curl_headers.png`

Observation: - **Server:** [e.g., nginx/1.18.0] - **X-Powered-By:** [e.g., PHP/7.4.3] - **Security Headers Present:** [e.g., Strict-Transport-Security] - **Security Headers Missing:** [e.g., X-Frame-Options, Content-Security-Policy]

4.5 Wafw00f – WAF Detection

Screenshot: `06_wafw00f.png`

Observation: - **WAF Detected:** [Yes / No] - **WAF Type:** [e.g., Cloudflare, ModSecurity, AWS WAF]

4.6 DNSRecon – DNS Enumeration

Screenshot: `07_dnsrecon.png`

Observation: - **A Records:** [List IPs] - **MX Records:** [List] - **TXT Records:** [e.g., SPF, DMARC] - **Subdomains Found:** [e.g., blog.networkwalks.com, mail.networkwalks.com]

5. Findings – Network Scanning (Local LAN)

5.1 Ping Scan (Host Discovery)

Screenshot: `01_ping_scan.png`

Target: [e.g., 192.168.1.0/24]

Observation: - **Live Hosts Found:** [Number] - **Host IPs:** [List IPs] - **MAC Addresses:** [List if available]

5.2 Quick Scan

Screenshot: `02_quick_scan.png`

Target: [e.g., 192.168.1.1]

Observation: - **Open Ports:** [e.g., 80, 443, 53] - **Services:** [e.g., http, https, domain]

5.3 Regular Scan

Screenshot: `03_regular_scan.png`

Target: [Same IP]

Observation: - **Open Ports:** [List] - **Service Versions:** [e.g., Apache httpd 2.4.41]

5.4 Intense Scan

Screenshot: `04_intense_scan.png`

Target: [Same IP]

Observation: - **Open Ports:** [List] - **Service Versions:** [List] - **OS Detection:** [e.g., Linux 4.x] - **Additional Script Output:** [e.g., HTTP title, SSH host key]

Analysis: The intense scan provided the most comprehensive view, including OS detection and script output. The target appears to be running [OS].

5.5 Network Topology

Screenshot: `05_topology.png`

6. Risk Analysis & Recommendations

#	Finding	Risk Level	Recommendation
1	[e.g., Outdated server version (nginx/1.18.0)]	Medium	Update to the latest stable version to mitigate known vulnerabilities.
2	[e.g., Missing HSTS header]	Low	Implement `Strict-Transport-Security` to prevent downgrade attacks.
3	[e.g., Open port 22 (SSH) on local host]	Medium	Enforce key-based authentication and disable root login.
4	[e.g., Open port 80 (HTTP) on local host]	Low	Redirect all HTTP traffic to HTTPS.
5	[e.g., WAF detected (Cloudflare)]	Informational	Positive finding — WAF provides additional protection.
6	[e.g., SPF/DMARC records present]	Informational	Positive finding — email security is configured.

Note: Risk levels are assessed as Low, Medium, High, or Informational based on the context of a passive footprinting and local lab scanning engagement. No exploitation was performed to confirm exploitability.

7. Conclusion

This project successfully demonstrated the reconnaissance and scanning phases of a penetration test. The footprinting phase revealed significant publicly available information about `networkwalks.com`, including its technology stack, DNS infrastructure, and WAF protection. The network scanning phase identified live hosts on the local lab network and mapped open ports and services on a target host.

Key takeaways: - Publicly available data, when aggregated, provides a detailed picture of an organization's attack surface. - Network scanning is essential for understanding what services are exposed and potentially vulnerable. - Proper authorization and scope definition are critical before any testing begins.

All activities were conducted ethically and within the authorized scope. No exploitation or disruptive testing was performed.